

ICST HOA Routing Checklist

ICST Ambisonics Plugins for REAPER · Institute for Computer Music and Sound Technology, ZHdK

BEFORE YOU START

- Open the REAPER project and save it under a new name.
- Verify audio device, sample rate (48 kHz), and buffer size.
- Confirm ICST Ambisonics Plugins v3.1+ are installed.

CHANNEL COUNT

- Set the Master track to 64 channels.
- Set all HOA-relevant tracks to 64 channels.
- Match the decoder output count to the loudspeaker setup.

SIGNAL FLOW

- Sources feed a clearly named HOA bus.
- The HOA bus feeds the decoder track.
- Optional binaural path branches from the same HOA bus.
- No accidental direct source-to-Master routes.

PLUGIN PLACEMENT

- Source tracks: ICST MonoEncoder or ICST MultiEncoder.
- Decoder track: AmbiDecoder or MultiDecoder plugin.
- Correct loudspeaker preset loaded in the decoder.

QUICK TEST

- Solo a single source and confirm level on the HOA bus.
- Move encoder position — verify expected spatial change.
- Decoder outputs reach the correct speakers.
- If binaural monitoring active: check headphone path separately.

BEFORE RECORDING / EXPORT

- Run a short test signal end-to-end.
- Save a versioned project file.
- Document export format and channel ordering.

■ Common Mistakes to Avoid

- *One track in the HOA path not set to 64 channels.*
- *Decoder placed on Master instead of after the HOA bus.*
- *Wrong loudspeaker preset loaded.*
- *Sources still sending directly to Master.*
- *Monitoring routed through the wrong output path.*